

Faculty Of Engineering and Technology

Electrical and Computer Engineering Department

ENEE 4113, Communication Laboratory

Double-side and Single-side Band Modulation

Pre-Lab Experiment #2

Prepared by:

Shadi Ataya, 1230159.

Instructor: Dr. Nofal Nofal

Teacher Assistant: Eng. Hazem Awaysa

Section: 5.

Date: 12|Oct|2025

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Message Signal

m(t) = 0.85 cos (2π (1000) t).

Block Diagram:

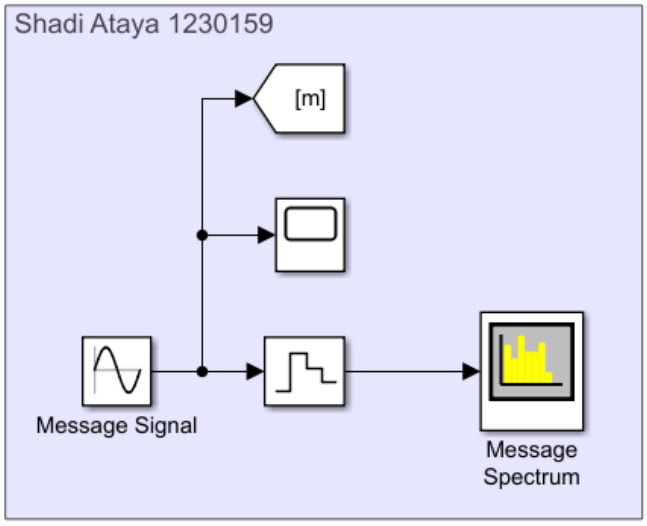


Figure 1 Message signal block diagram

Parameters:

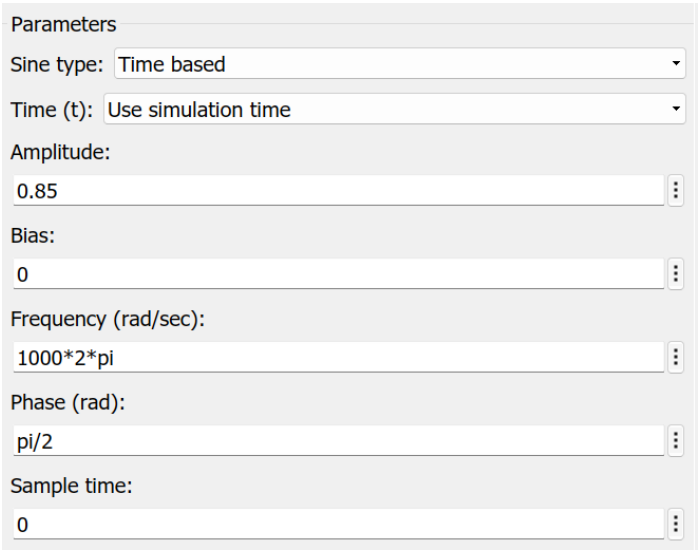


Figure 2 Message signal parameters

Message signal in time domain:

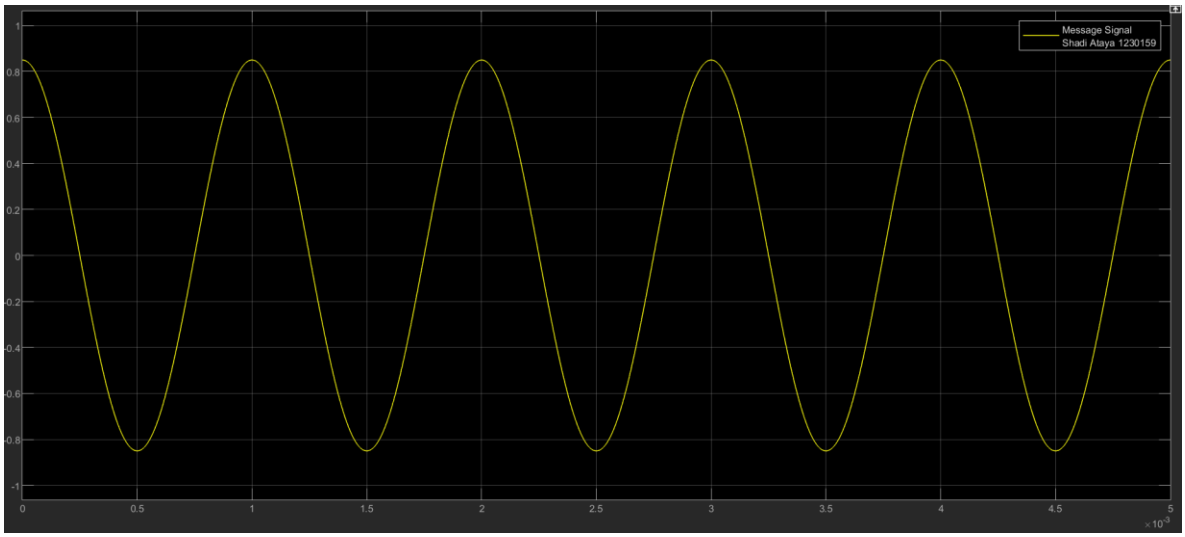


Figure 3 Message signal in time domain

Message signal in frequency domain

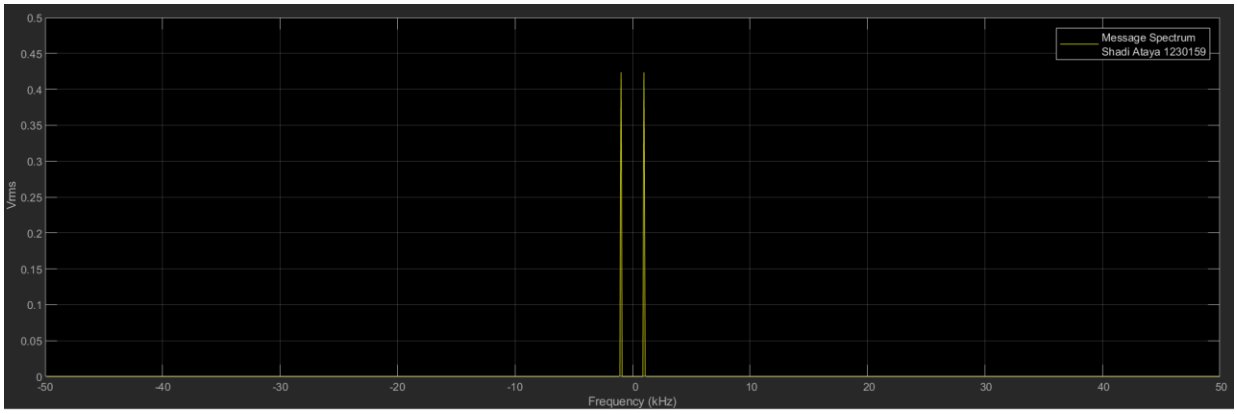


Figure 4 Message signal in frequency domain

Carrier Signal

c(t) = 1cos (2π (15k) t).

Block Diagram:

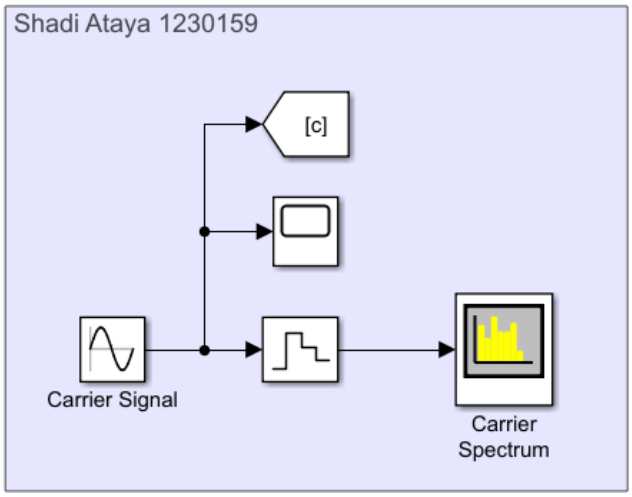


Figure 5 Carrier signal block diagram

Parameters:

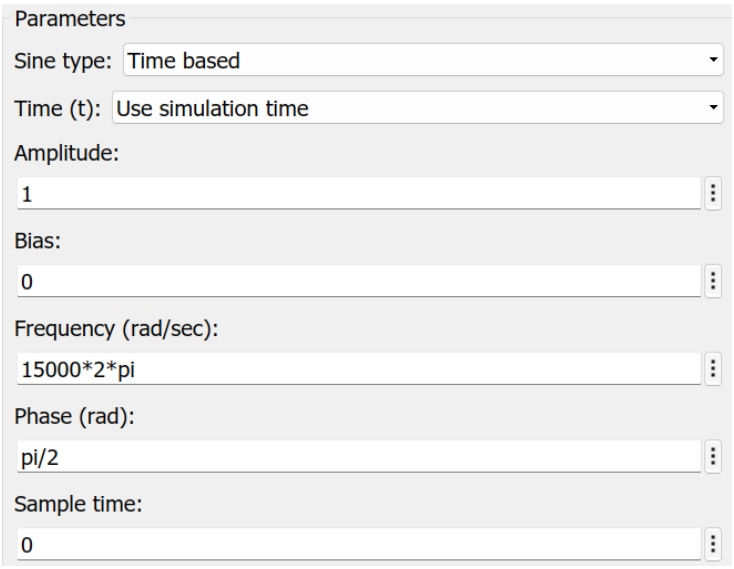


Figure 6 Carrier signal parameters

Carrier signal in time domain:

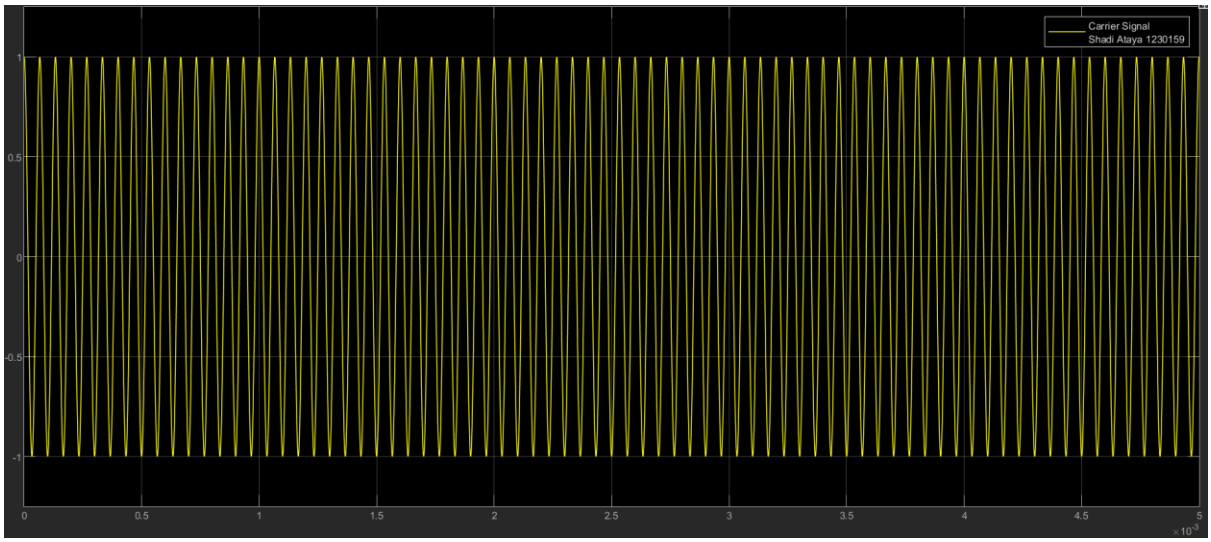


Figure 7 Carrier signal in time domain

Carrier signal in frequency domain

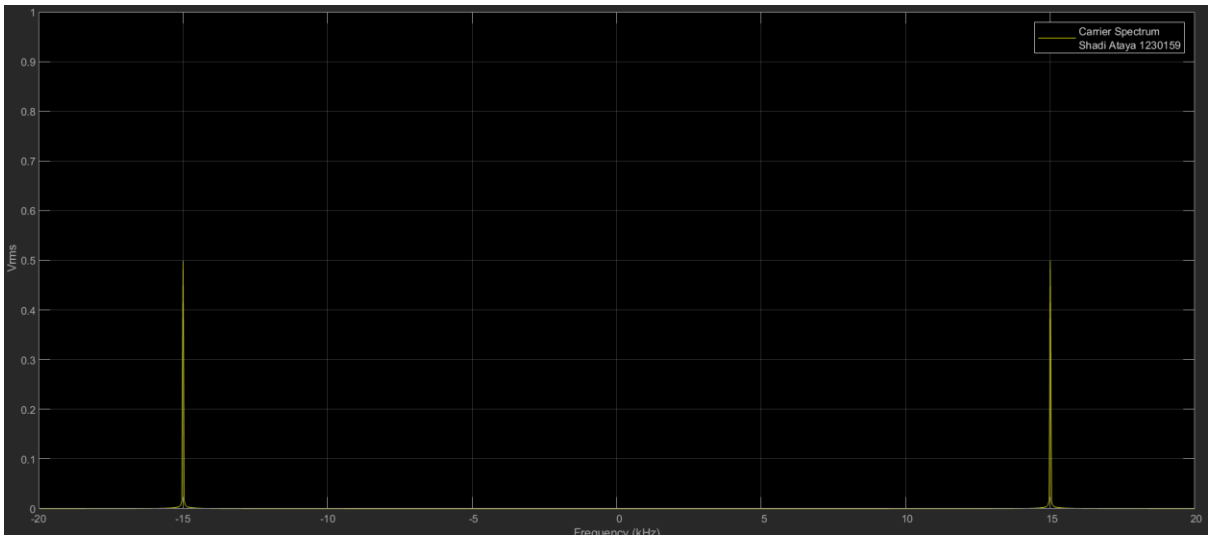


Figure 8 Carrier signal in frequency domain

DSB-SC

Part 1: Modulation

Block Diagram:

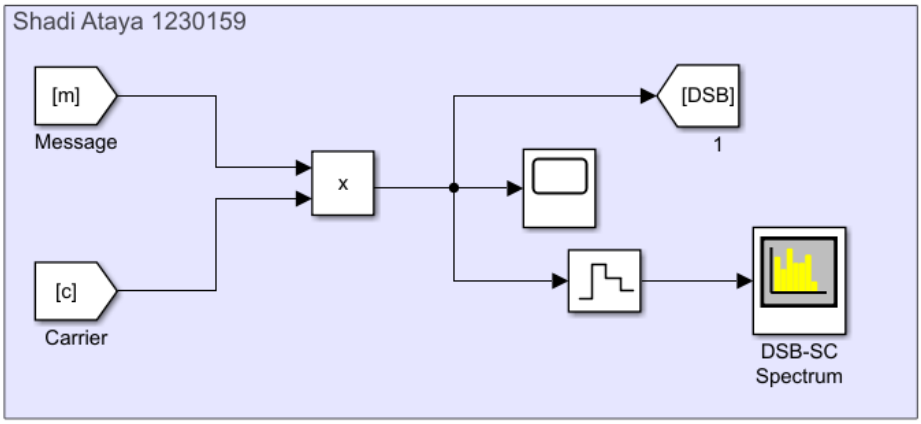


Figure 9 DSB-SC modulation block diagram

Modulated signal in time domain

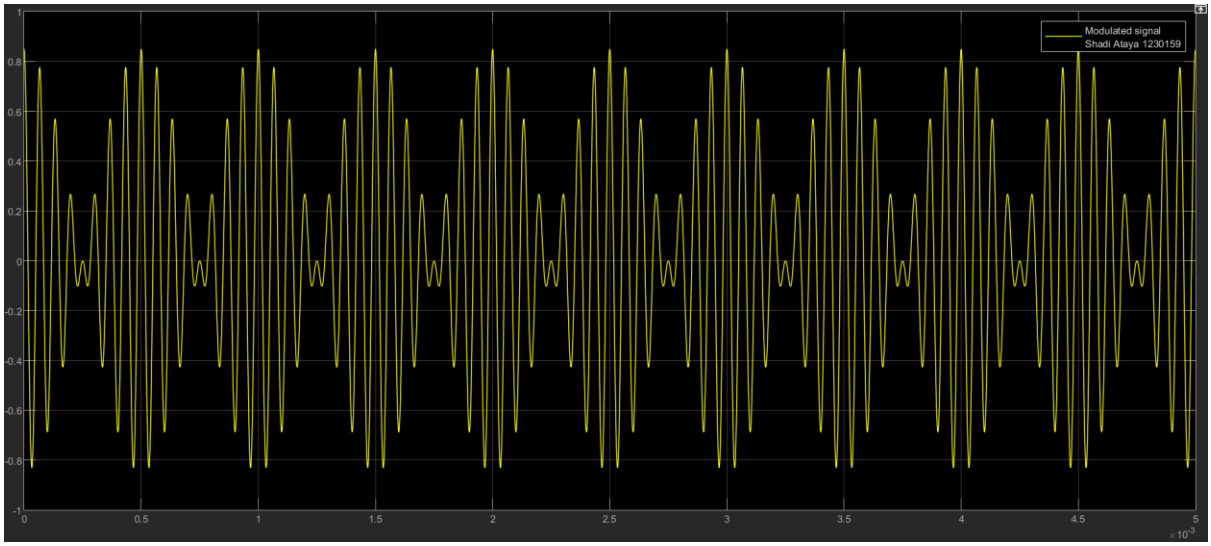


Figure 10 Modulated signal in time domain

Modulated signal in frequency domain

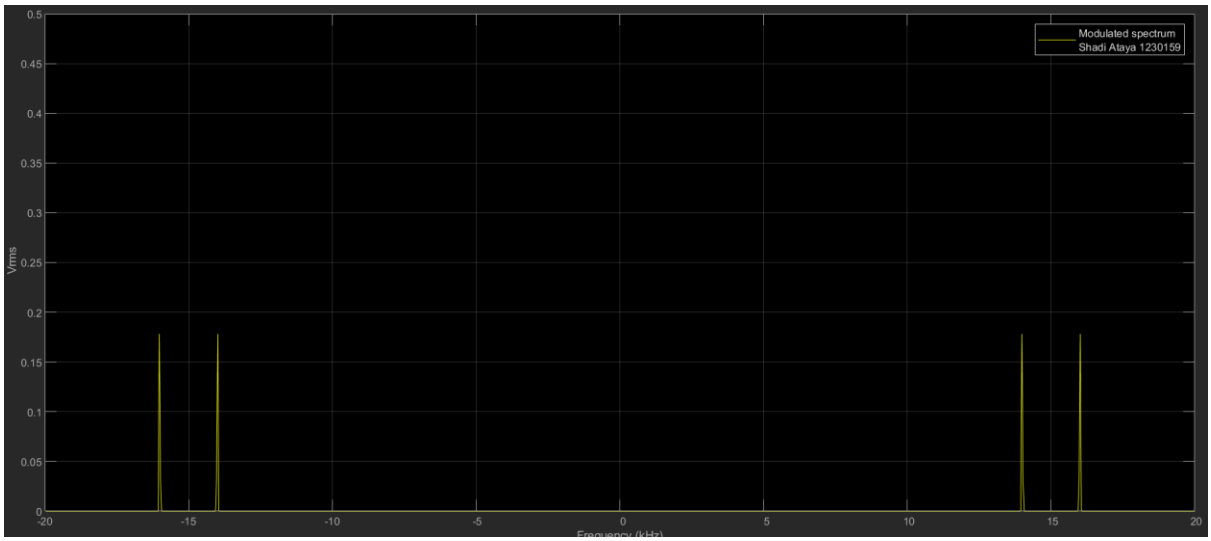


Figure 11 Modulated signal in frequency domain

Part 2: Demodulation

Block Diagram:

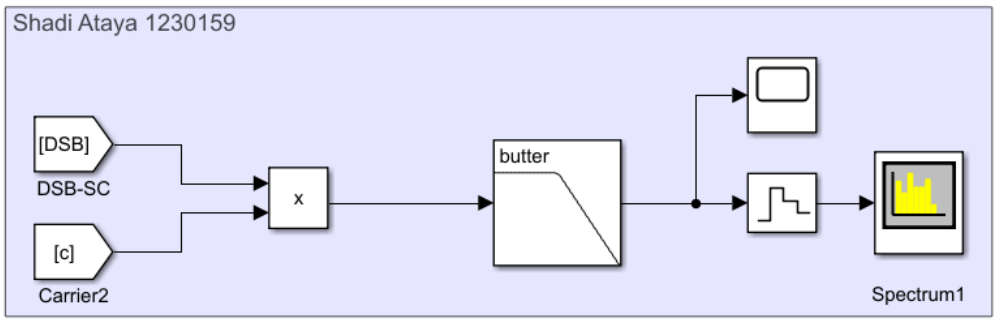


Figure 12 DSB-SC demodulation block diagram

Demodulated signal in time domain

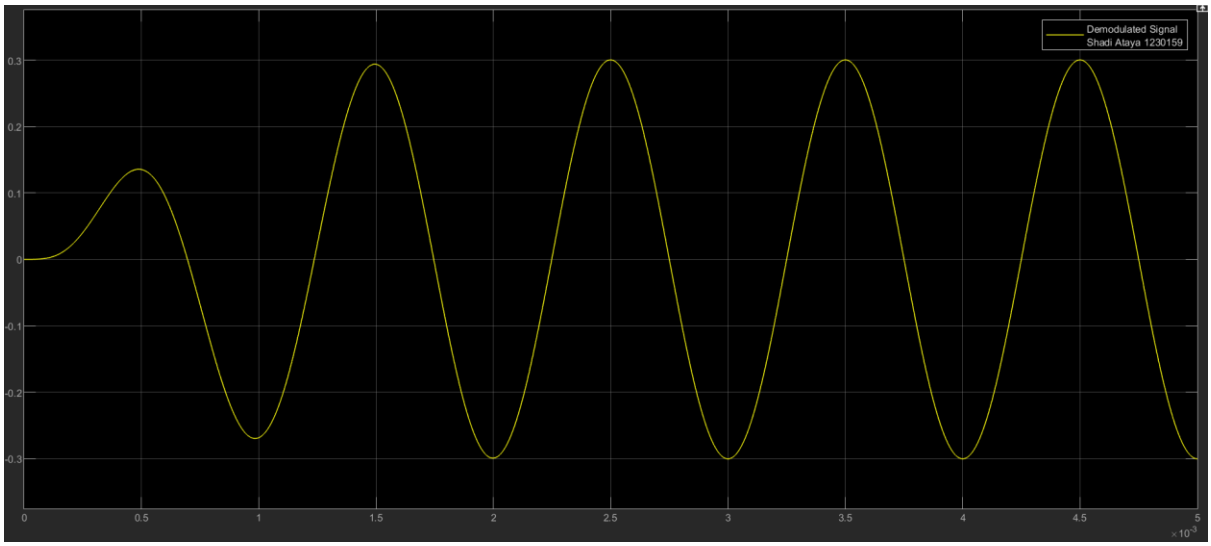


Figure 13 Demodulated signal in time domain

Demodulated signal in frequency domain

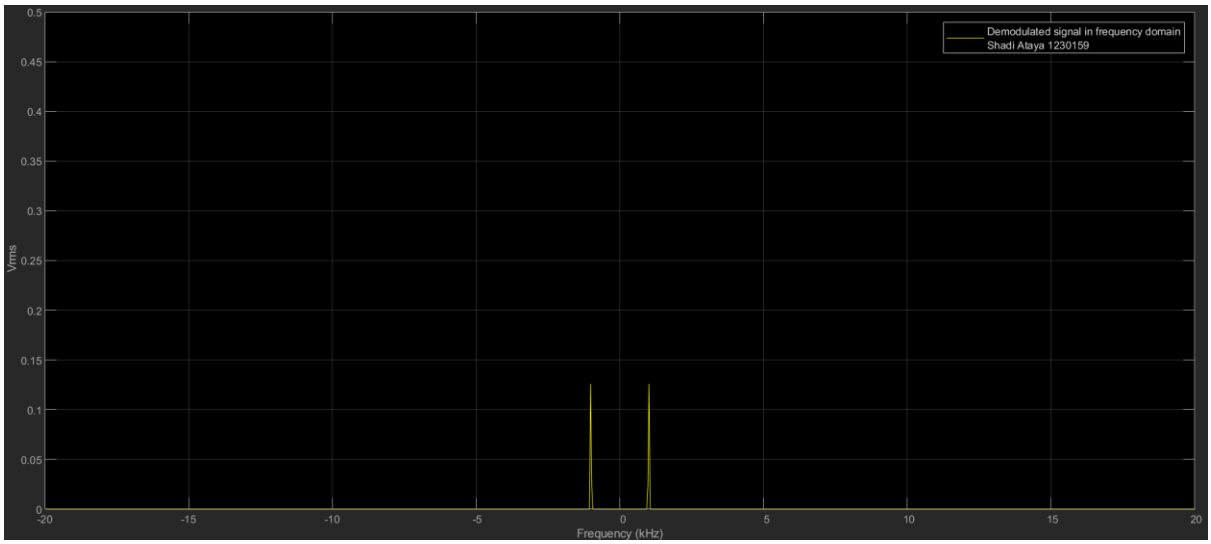


Figure 14 Demodulated signal in frequency domain

SSB-SC (First Method)

Part 1: Modulation

Block Diagram:

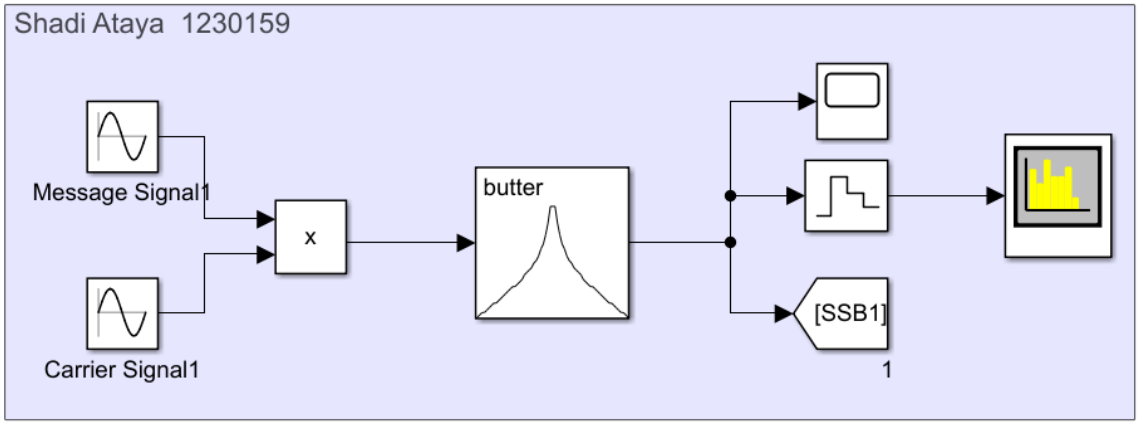


Figure 15 SSB modulation block diagram

Modulated signal in time domain

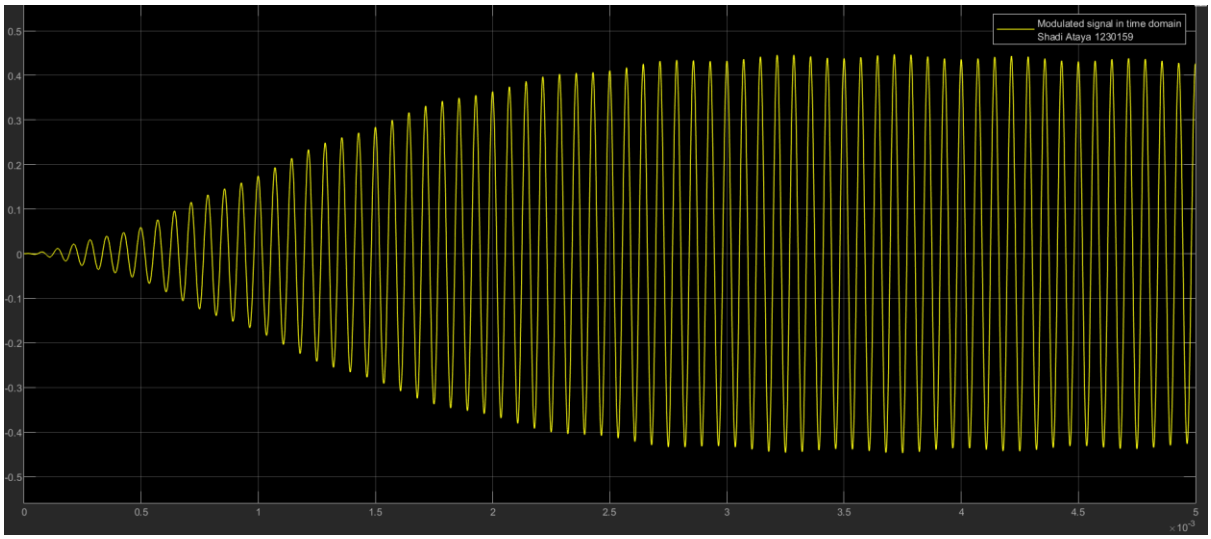


Figure 16 Modulated signal in time domain

Modulated signal in frequency domain

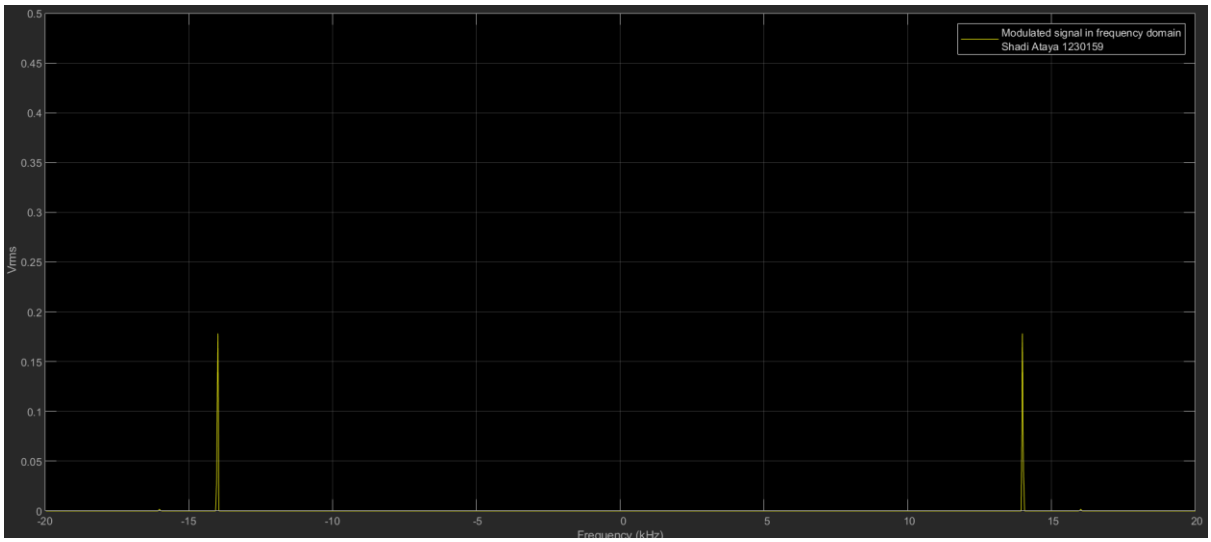


Figure 17 Modulated signal in frequency domain

Part 2: Demodulation

Block Diagram:

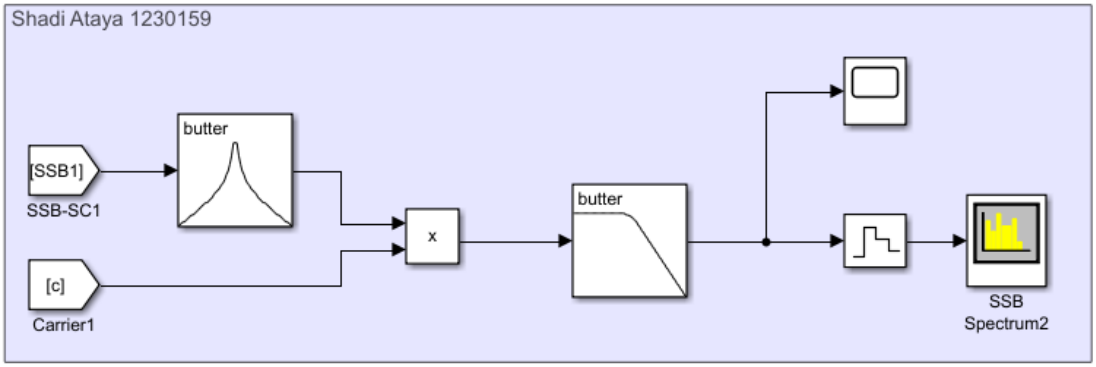


Figure 18 SSB demodulation block diagram

Demodulated signal in time domain

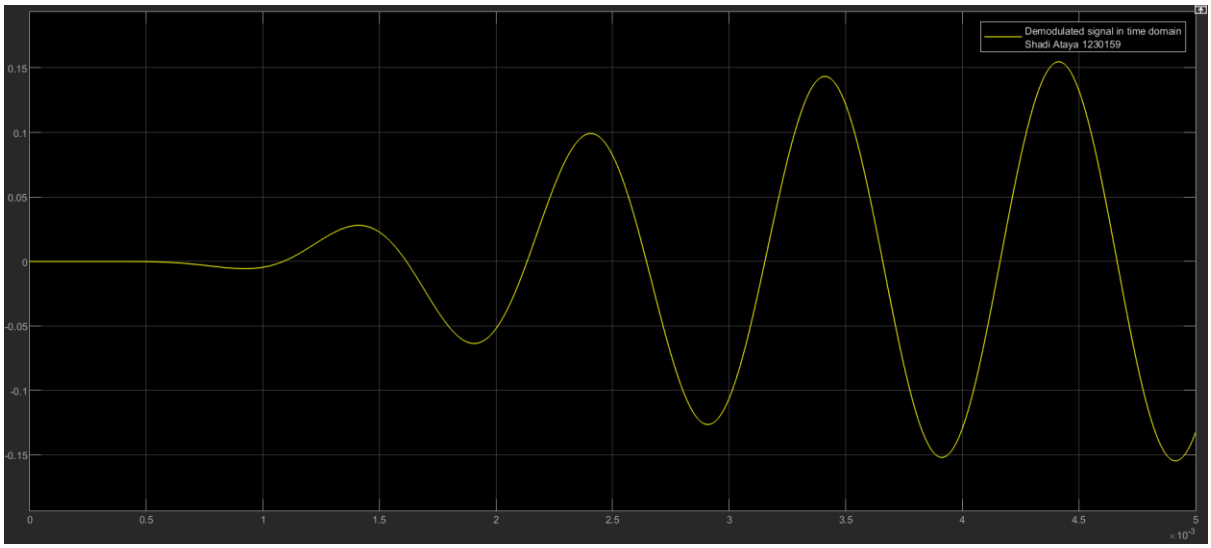


Figure 19 Demodulated signal in time domain

Demodulated signal in frequency domain

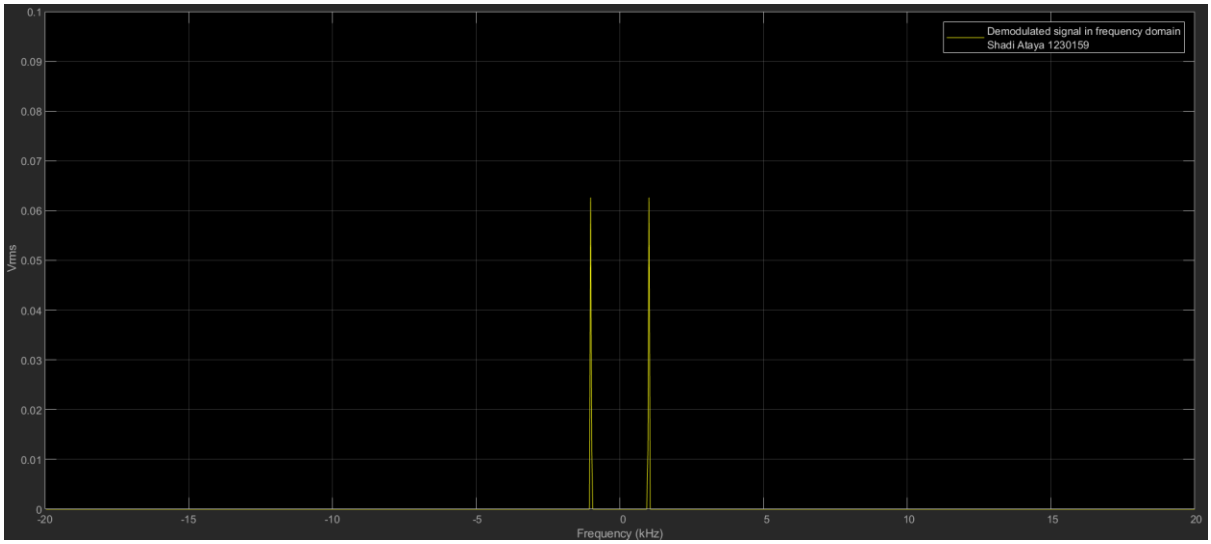


Figure 20 Demodulated signal in frequency domain

SSB-SC (Second Method)

Part 1: Modulation

Block Diagram:

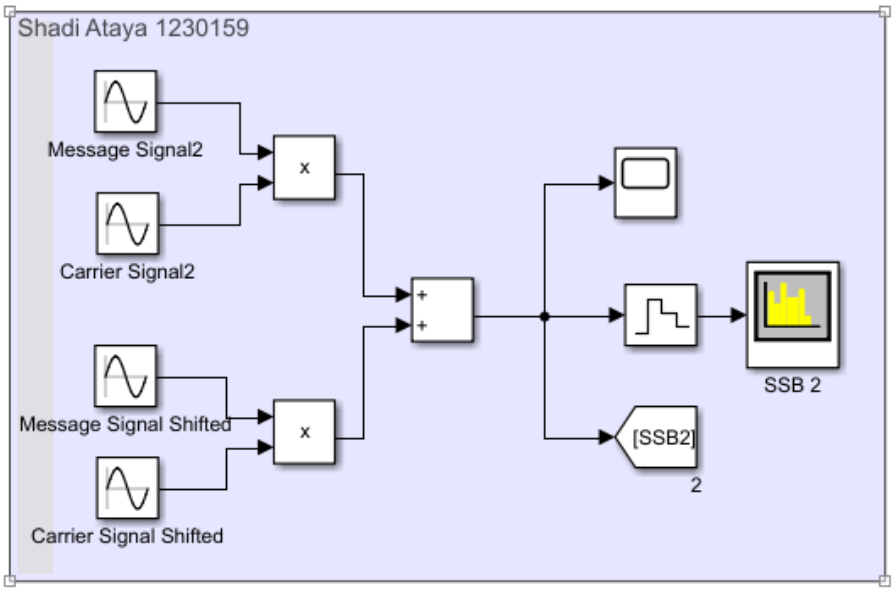


Figure 21 SSB-2 modulation block diagram

Modulated signal in time domain

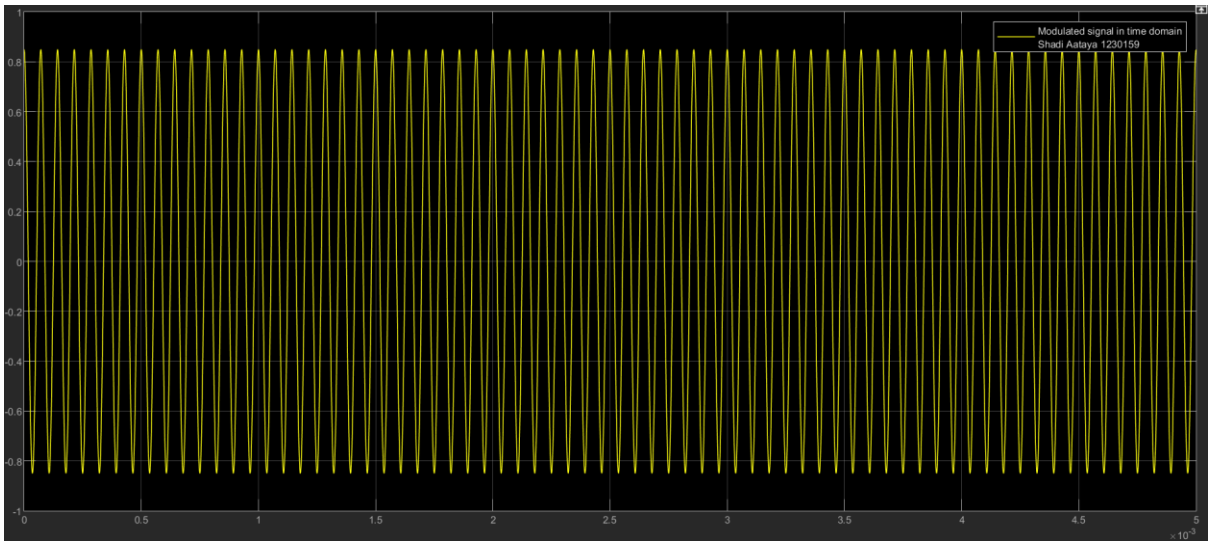


Figure 22 Modulated signal in time domain

Modulated signal in frequency domain

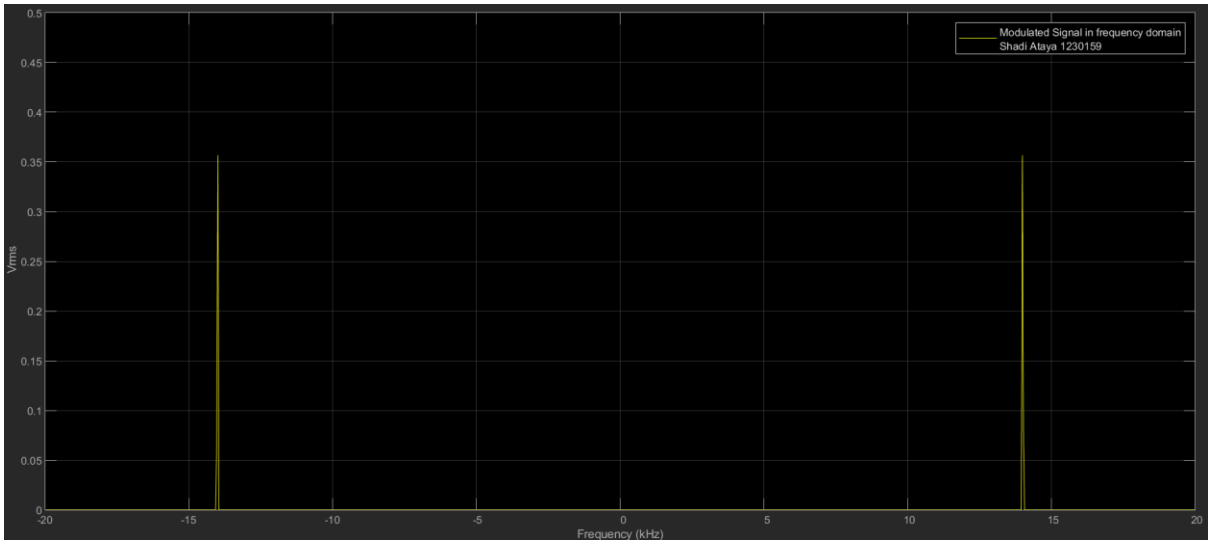


Figure 23 Modulated signal in frequency domain

Part 2: Demodulation

Block Diagram:

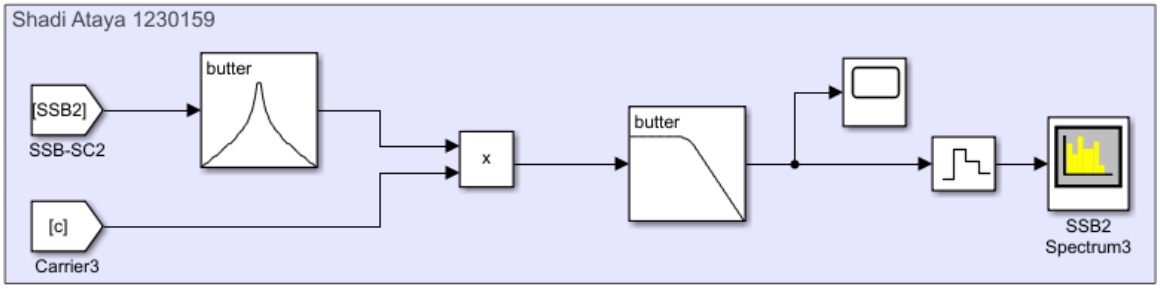


Figure 24 SSB-2 demodulation block diagram

Demodulated signal in time domain

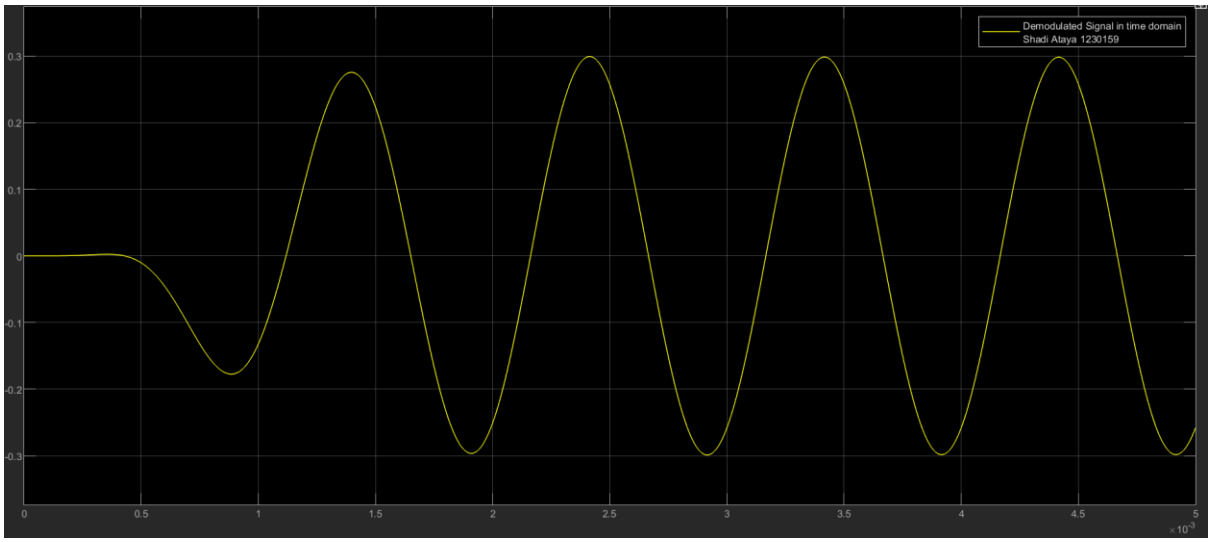


Figure 25 Demodulated signal in time domain

Demodulated signal in frequency domain

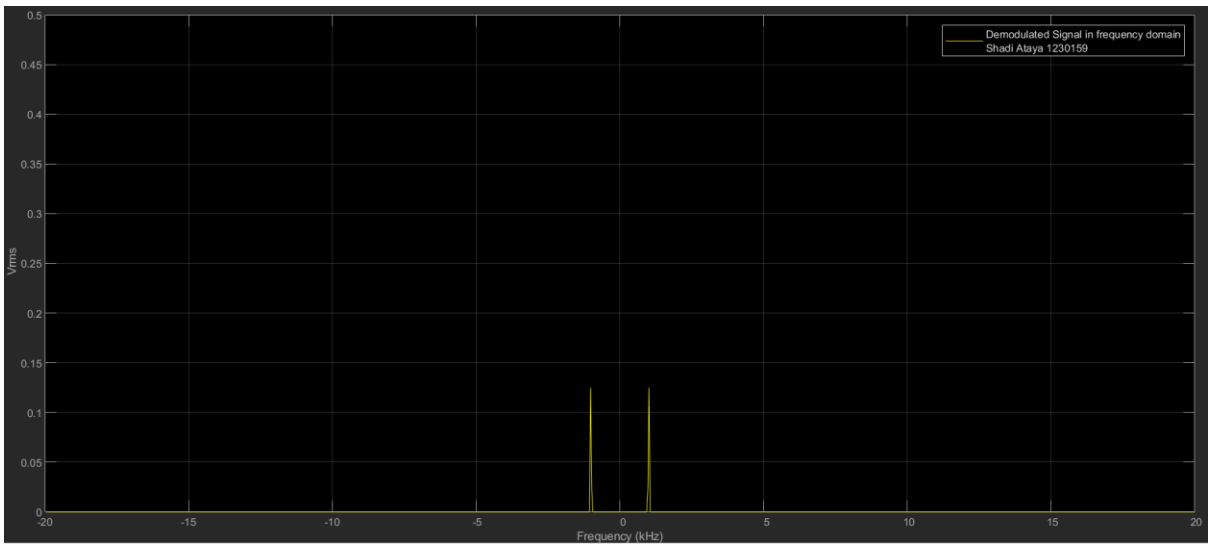


Figure 26 Demodulated signal in frequency domain